In class lab #4 for Numerical Analysis

Your name: _____

scan and submit before 23:00

Part I: Gaussian elimination (20 points)

Problem 1. Consider the system

$$\begin{aligned} 2x_1 - 2x_2 - x_3 &= -2 \\ 4x_1 + x_2 - 2x_3 &= 1 \\ -2x_1 + x_2 - x_3 &= -3 \end{aligned}$$

Step 1 (5 points) Two steps to eliminate column 1:

$$\begin{bmatrix} & & & | & \\ & & & | & \\ & & & | & \end{bmatrix} \longrightarrow \begin{array}{c}\text{subtract } ___ \times \text{row1} \\ \text{from row2}\end{array} \longrightarrow \left[\begin{array}{cccc|c} ___ & ___ & ___ & | & ___ \\ 0 & ___ & ___ & | & ___ \\ ___ & ___ & ___ & | & ___ \end{array}\right]$$

$$\longrightarrow \begin{array}{c}\text{subtract } ___ \times \text{row1} \\ \text{from row3}\end{array} \longrightarrow \left[\begin{array}{cccc|c} ___ & ___ & ___ & | & ___ \\ 0 & ___ & ___ & | & ___ \\ 0 & ___ & ___ & | & ___ \end{array}\right]$$

and one more step to eliminate column 2:

$$\left[\begin{array}{cccc|c} ___ & ___ & ___ & | & ___ \\ 0 & ___ & ___ & | & ___ \\ 0 & ___ & ___ & | & ___ \end{array}\right] \longrightarrow \begin{array}{c}\text{subtract } ___ \times \text{row2} \\ \text{from row3}\end{array} \longrightarrow \left[\begin{array}{cccc|c} ___ & ___ & ___ & | & ___ \\ 0 & ___ & ___ & | & ___ \\ 0 & 0 & ___ & | & ___ \end{array}\right]$$

Step 2 (5 points) To do the back substitution:

From the last equation, we have

$$x_3 = \frac{b_3}{a_{33}} =$$

From the second equation, we have

$$x_2 = \frac{b_2 - a_{23}x_3}{a_{22}} =$$

From the first equation, we have

$$x_1 = \frac{b_1 - a_{13}x_3 - a_{12}x_2}{a_{11}} =$$

Step 3 (5 points) Click on New to create a new document and type the code below. Once you are done, save it by the name `gaussian_elimination.m`.

```
%Program 2.1 Gaussian elimination
function [x,n_operations] = gaussian_elimination(a,b)

n = length(b);
% Step 1: to eliminate the entries in the lower triangular.
for j = 1 : n-1
if abs( a(j,j) )<eps; error('zero pivot encountered'); end
for i = j + 1 : n
multi = a(i,j)/a(j,j);
```

```
for k = j+1 : n
a(i,k) = a(i,k) - multi*a(j,k);
end
b(i) = b(i) - multi*b(j);
end
end
% Step 2: to do back substitution.
for i = n : -1 : 1
for j = i+1 : n
b(i) = b(i) - a(i,j)*x(j);
end
x(i) = b(i)/a(i,i);
end

% output the number of operations of Gaussian elimination
n_operations = 2/3*n*n*n+1/2*n*n-7/6*n+n*n;
```

Step 4 (5 points) Type

```
>> a= [ 2,-2 ,-1 ; 4 , 1 , -2 ; -2,1,-1 ]

a =


---   ---   ---
---   ---   ---
---   ---   ---

>> b=[ -2;1;-3 ]

b =


---
---
---

>> [x,n_operations] = gaussian_elimination(a,b)

x =


------    ------   ------


n_operations =

---
```

Part II: The LU factorization (40 points)

Problem 2. Example: To find the LU factorization for the matrix

$$\begin{bmatrix} 1 & 1 \\ 3 & -4 \end{bmatrix}$$

To eliminate a_{21},

$$\begin{bmatrix} 1 & 1 \\ 3 & -4 \end{bmatrix} \longrightarrow \begin{array}{c}\text{subtract} \quad \mathbf{3} \times \text{row1} \\ \text{from row2}\end{array} \longrightarrow \begin{bmatrix} 1 & 1 \\ 0 & -7 \end{bmatrix} = U.$$

Then the lower triangular

$$L = \begin{bmatrix} 1 & 0 \\ \mathbf{3} & 1 \end{bmatrix}$$

Step 1 (5 points) To find the LU factorization of A.

Similar to the above example, the elimination steps proceed as Step 1.1:

$$\begin{bmatrix} 2 & -2 & -1 \\ 4 & 1 & -2 \\ -2 & 1 & -1 \end{bmatrix} \longrightarrow \begin{array}{c}\text{subtract} \quad ___ \times \text{row1} \\ \text{from row2}\end{array} \longrightarrow \begin{bmatrix} 2 & -2 & -1 \\ 0 & ___ & ___ \\ ___ & ___ & ___ \end{bmatrix}$$

$$\longrightarrow \begin{array}{c}\text{subtract} \quad ___ \times \text{row1} \\ \text{from row3}\end{array} \longrightarrow \begin{bmatrix} 2 & -2 & -1 \\ 0 & ___ & ___ \\ 0 & ___ & ___ \end{bmatrix}$$

$$\longrightarrow \begin{array}{c}\text{subtract} \quad ___ \times \text{row2} \\ \text{from row3}\end{array} \longrightarrow \begin{bmatrix} 2 & -2 & -1 \\ 0 & ___ & ___ \\ 0 & 0 & ___ \end{bmatrix} = U.$$

Then the lower triangular L matrix is

$$\begin{bmatrix} 1 & 0 & 0 \\ ___ & 1 & 0 \\ ___ & ___ & 1 \end{bmatrix}$$

Step 2 (5 points) As in Step 2.1, we put 1's on the main diagonal and the multipliers in the lower triangular—in the specific places they were used for elimination. We write the following MATLAB function and save it by the name `LU_factorization.m`:

```
function [L,U,n_operations] = LU_factorization(a)

n = length(a)
L = eye(n);
% Step 1: to eliminate the entries in the lower triangular.
for j = 1 : n-1
if abs( a(j,j) )<eps; error('zero pivot encountered'); end
for i = j + 1 : n
L(i,j) = a(i,j)/a(j,j);
for k = j+1 : n
a(i,k) = a(i,k) - L(i,j)*a(j,k);
end
end
end

U = zeros(n);
for i = 1 : n
```

```
for j = 1 : n
if(i<=j)
U(i,j) = a(i,j);
end
end
end

% output the number of operations of getting L and U
n_operations = 2/3*n*n*n-1/2*n*n-1/6*n;
```

Step 3 (5 points) Type

```
>> [L,U,n_operations] = LU_factorization(a)

L =


---   ---   ---
---   ---   ---
---   ---   ---


U =


---   ---  ---
---   ---  ---
---   ---   ---


n_operations =


---
```

Step 4 (5 points) Once L and U are known, the problem $Ax = b$ can be written as $LUx = b$. Define a new "auxiliary" vector $c = Ux$. Then back substitution is a two-step procedure:

(a) Solve $Lc = b$ for c.

(b) Solve $Ux = c$ for x. The procedure has been presented in Step 2 of program 2.1.

We write the following MATLAB function and save it by the name `LU_forward_back.m`:

```
function [x,n_operations] = LU_forward_back(L,U,b)

n = length(b);

% Step 1: to do a forward substituion.
for i = 1 : 1 : n
for j = 1 : i-1
b(i) = b(i) - L(i,j)*c(j);
end
c(i) = b(i)/L(i,i);
end
% Step 2: to do a back substitution.
for i = n : -1 : 1
for j = i+1 : n
c(i) = c(i) - U(i,j)*x(j);
end
```

```
x(i) = c(i)/U(i,i);
end

% output the number of operations of forward and backward substitution
n_operations = 2*n*n;
```

Step 5 (5 points) Type

```
>> [x,n_operations] = LU_forward_back(L,U,b)

x =

-----    ------   -------


n_operations =

---
```

Problem 3. (15 points) Here is the reason why we have to learn "how" of the LU factorization. Now, suppose that we need to solve four problems with the same A and different b.

$$Ax = b_1$$
$$Ax = b_2$$
$$Ax = b_3$$
$$Ax = b_4$$

By Gaussian elimination We have to perform an elimination to get U for each prolem.

By LU approach We perform only one elimination "$A = LU$" to get U and do a forward and back substitution ($Lc = b_1, Ux = c$) for the first problem with b_1. For the later problems (that is, 2nd, 3rd, 4th equations) with $b_i, i = 2, 3, 4$, we only need to do a forward and back substitution ($Lc = b_i, Ux = c$).

Please count the operations of Gaussian elimination and LU approach for solving 4 problems with the same $A \in \mathbb{R}^{3\times 3}$ and different $b_i \in \mathbb{R}^3$.

(1)

$$\text{Operations of Gaussian elimination} = 4 \times _____ = _____.$$

(Hint: Part I, step 3)

(2)

$$\text{Operations of LU approach} = \underset{\text{for } Ax=b_1}{____ + ____} \quad \underset{\text{for } Ax=b_2}{+____} \quad \underset{\text{for } Ax=b_3}{+____} \quad \underset{\text{for } Ax=b_4}{+____ = _____}.$$

(Hint: Part II, step 2 and step 4)

Part III: Gaussian elimination with partial pivoting (40 points)

The form of Gaussian elimination considered so far is often called "naive," because of two serious difficulties: encountering a zero pivot and swamping. For a nonsingular matrix, both can be avoided with an improved algorithm. The key to this improvement is an efficient protocol for **exchanging rows** of the coefficient matrix, called **partial pivoting**.

Apply Gaussian elimination with partial pivoting to solve the system

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array}\right].$$

We do the following procedure to eliminate a_{21}, a_{31}, a_{32}.

To eliminate a_{21}, a_{31},

Step 1. To locate the largest entry of the first column and we swap its row with the pivot row.

- e.g., If $|a_{21}|$ is the largest entry among $|a_{11}|, |a_{21}|, |a_{31}|$, we swap row 2 with row 1.

Step 2. To do Gaussian elimination for eliminating the new a_{21}, a_{31}.

To eliminate a_{32},

Step 1. To locate the largest entry between $|a_{22}|$ and $|a_{32}|$.

- e.g., If $|a_{32}|$ is the largest entry between $|a_{22}|$ and $|a_{32}|$, we swap row 3 with row 2.

Step 2. To do Gaussian elimination for eliminating the new a_{32}.

Problem 4. (20 points) **Complete the code in the box** to make it a Gaussian elimination with partial pivoting function. Click on New to create a new document and type the code. Once you are done, save it by the name `gaussian_elimination_partial_pivoting.m`.

```
function [x,n_operations] = gaussian_elimination_partial_pivoting(a,b)

n = length(b);
% Step 1: to eliminate the entries in the lower triangular.
for j = 1 : n-1
-------------------------------------------------------------------------
|%  Partial pivoting process                                            |
|                                                                       |
|                                                                       |
|                                                                       |
|                                                                       |
|                                                                       |
|                                                                       |
|                                                                       |
|                                                                       |
|                                                                       |
|                                                                       |
-------------------------------------------------------------------------
if abs( a(j,j) )<eps; error('zero pivot encountered'); end
for i = j + 1 : n
multi = a(i,j)/a(j,j);
for k = j+1 : n
a(i,k) = a(i,k) - multi*a(j,k);
end
b(i) = b(i) - multi*b(j);
end
```

```
end
% Step 2: to do back substitution.
for i = n : -1 : 1
for j = i+1 : n
b(i) = b(i) - a(i,j)*x(j);
end
x(i) = b(i)/a(i,i);
end

% output the number of operations of Gaussian elimination with partial pivoting
n_operations = 2/3*n*n*n+1/2*n*n-7/6*n+n*n;
```

Problem 5. (20 points) Use the `gaussian_elimination.m` and `gaussian_elimination_partial_pivoting.m` to solve the system

$$\begin{bmatrix} 6 & 1 & & & & & \\ 8 & 6 & 1 & & & & \\ & 8 & 6 & 1 & & & \\ & & 8 & 6 & 1 & & \\ & & & \ddots & \ddots & \ddots & \\ & & & & 8 & 6 & 1 \\ & & & & & 8 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ \vdots \\ x_{85} \\ x_{86} \end{bmatrix} = \begin{bmatrix} 7 \\ 15 \\ 15 \\ 15 \\ \vdots \\ 15 \\ 14 \end{bmatrix}$$

respectively.

(a) Plot two estimated solutions $\mathbf{x}_1, \mathbf{x}_2$ and the exact solution $\mathbf{x}^* =$ ones(86,1) in one graph.

(b) Use $\mathbf{e}_i = ||\mathbf{x}_i - \mathbf{x}^*||_2$ to calculate the errors, $i = 1, 2$.